

Interfacial and Piezoelectric Physics of PVDF–Metal Oxide Nanocomposites for Energy Harvesting

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Abstract

In PVDF–metal oxide nanocomposites, the gain in harvested energy comes mainly from interfacial nucleation of polar phases, not from the filler's own piezoelectricity. We build a unified, first-order model linking four things: phase quantification, interphase geometry, dielectric mixing and the piezoelectric coupling of 0–3 composites. Device output under a matched resistive load closes the chain. The model is implemented as an open, tested Python toolkit (`pvdf_nano`).

Three quantitative results follow. At 5 vol% loading with a 5 nm interphase, the interphase occupies about 11% of a 20 nm-particle composite but only 0.15% of a 1 μm -particle one. A BaTiO₃ particle in PVDF sees only about 2% of the applied field (local field coefficient $L_E \approx 0.021$), so it contributes little piezoelectric response directly. Because PVDF has a negative d_{33} and oxide ceramics a positive one, co-poled fillers reduce the composite response; antiparallel poling makes the two add.

At 10 vol% loading and equal polar-phase content, a ZnO-filled film reaches a transduction figure of merit $d_{33} \cdot g_{33}$ of $5.98 \times 10^{-12} \text{ m}^2/\text{N}$. A BaTiO₃-filled film reaches $4.38 \times 10^{-12} \text{ m}^2/\text{N}$ and a neat film $2.55 \times 10^{-12} \text{ m}^2/\text{N}$. The lower permittivity of ZnO explains its advantage. We conclude with design rules: nanoscale particles, surface chemistry that fixes surface charge, low permittivity mismatch, and optimisation of $d_{33} \cdot g_{33}$ rather than d_{33} .

Keywords: PVDF; β -phase; nanocomposite; interphase; Maxwell–Wagner–Sillars polarisation; piezoelectric nanogenerator; energy harvesting

1. Introduction

Wearable sensors, implants and distributed IoT nodes need milliwatt- to microwatt-level power without battery replacement. Mechanical energy from body motion, vibration, wind and flow is widely available at 1–100 Hz. Piezoelectric nanogenerators (PENGs) convert it directly into electrical charge [11].

Poly(vinylidene fluoride) (PVDF) suits this role. Its $-\text{CH}_2\text{--CF}_2-$ repeat unit carries a dipole of about 2.1 D. It is flexible, chemically stable and easy to process into films and fibres. Its piezoelectric coefficient is modest ($|d_{33}| \approx 20\text{--}35 \text{ pC/N}$), but its permittivity is low ($\epsilon_r \approx 10\text{--}12$). This gives a voltage coefficient $|g_{33}|$ of $0.2\text{--}0.33 \text{ V}\cdot\text{m/N}$, about ten times that of PZT [1, 3].

PVDF's response is limited by three factors: the degree of crystallinity X_c , the fraction of crystals in a polar phase $F(\beta)$, and how completely those crystals are poled. Melt-processed PVDF crystallises mostly in the non-polar α -phase. Metal-oxide nanoparticles such as ZnO, BaTiO₃, TiO₂ and Fe₃O₄ are widely added to raise $F(\beta)$. Many studies report d_{33} or output-voltage gains at loadings of a few weight percent [3].

The physics behind these gains is often mixed together in the literature. Studies conflate four mechanisms: nucleation of polar phases at the interface, the filler's own piezoelectricity, interfacial space charge, and triboelectric artefacts in device tests. Reports also frequently use the FTIR electroactive fraction as the β -fraction, even though the 840 cm^{-1} band includes the γ -phase [4].

This paper makes four contributions:

1. It assembles phase quantification, interphase geometry, dielectric mixing, 0–3 piezoelectric coupling and device electrical output into one consistent model chain.
2. It makes the sign of d_{33} explicit and shows how filler poling direction changes the composite response.
3. It quantifies why high-permittivity piezoceramic fillers couple poorly to PVDF and why low-permittivity fillers can deliver more energy.
4. It releases the model as a tested, open Python toolkit so the results can be reproduced and extended with measured data.

2. Polymorphism and phase quantification of PVDF

Only the polar β - and γ -phases contribute to piezoelectricity, so the harvesting-relevant quantity is the polar crystalline content $X_c \cdot F(\text{polar})$. Table 1 lists the three phases that matter and their signatures [1, 3, 4].

Table 1. Main PVDF polymorphs and their signatures (XRD for Cu K α radiation).

Phase	Chain conformation	Polarity	FTIR bands (cm^{-1})	XRD 2θ ($^\circ$)
α (form II)	TGTG'	Non-polar (antiparallel chains)	614, 763, 795, 976	17.7, 18.4, 19.9, 26.6

3. Interfacial physics

At nanometre particle sizes the interphase, not the particle, controls the composite. Its volume grows as the inverse of particle radius, and its structure sets chain conformation, charge distribution and mobility.

3.1 Interphase volume

For spheres of radius r carrying a shell of thickness t at filler volume fraction ϕ , the dilute interphase fraction is $\phi[(1 + t/r)^3 - 1]$. At higher loading the shells overlap. We use a random-overlap (Poisson) correction that keeps the interphase within the available matrix volume:

$$\phi_i = (1 - \phi) \left[1 - \exp\left\{-\frac{\phi \left[(1 + t/r)^3 - 1\right]}{1 - \phi}\right\}\right]$$

The specific interfacial area is $3\phi/r$. On a simple cubic lattice, the mean surface-to-surface spacing between particles of diameter d is $s = d[(\pi/6\phi)^{1/3} - 1]$.

3.2 Interphase structure

Tanaka's multi-core model describes the interphase as concentric layers [5]. Table 2 lists them with typical thicknesses.

Table 2. Layers of the multi-core interphase model.

Layer	Typical thickness	Physical character
Bonded	~1 nm	Chains tied to the surface by covalent, hydrogen or ionic bonds
Bound	2–9 nm	Ordered, immobilised chains; where conformational templating acts
Loose	Tens of nm	Altered mobility, free volume and crystallisation kinetics
Diffuse double layer	λ_D , nm to μm	Gouy–Chapman space charge overlapping the layers above

The extent of the diffuse layer is the Debye length, where n is the number density of each ion species of valence z :

$$\lambda_D = \sqrt{\frac{\epsilon_r \epsilon_0 k_B T}{2 n z^2 e^2}}$$

With ppm-level ionic impurities, λ_D in PVDF spans tens of nanometres to micrometres. Double layers of neighbouring particles therefore overlap at modest loading. Lewis argued that such interfaces dominate dielectric behaviour at nanometric dimensions [6].

3.3 Nucleation of polar phases

Five mechanisms stabilise β and γ conformations near oxide surfaces.

1. **Surface charge (ion–dipole).** A negative surface ($-\text{O}^-$, $-\text{OH}$) attracts the $-\text{CH}_2-$ side of the chain. A positive surface (under-coordinated Zn^{2+} , Ti^{4+} , Fe^{3+}) attracts $-\text{CF}_2-$. Either way, dipoles are pinned normal to the surface, which only all-trans or T3G sequences allow.
2. **Hydrogen bonding.** Surface $-\text{OH}$ groups form $\text{C}-\text{F}\cdots\text{H}-\text{O}$ bonds that anchor CF_2 groups.
3. **Lattice matching.** Surface spacings near the β (110)/(200) spacing (~ 0.43 nm) or the chain repeat (~ 0.256 nm) favour heterogeneous β nucleation.
4. **Confinement and stress.** Narrow gaps between particles restrict lamellar growth and favour extended chains, as mechanical stretching does.
5. **Altered kinetics.** Particles act as heterogeneous nuclei, raising crystallisation temperature and nucleation density.

The strength of mechanism 1 can be estimated. The field just outside a surface of charge density σ is $E = \sigma/(\epsilon_r \epsilon_0)$. The ratio of dipole alignment energy to thermal energy is

$$x = \frac{\mu E}{k_B T} = \frac{\mu \sigma}{\epsilon_r \epsilon_0 k_B T}, \quad \langle \cos \theta \rangle = \coth x - \frac{1}{x}$$

For $\sigma = 0.01$ C/m², $\epsilon_r = 10$ and $\mu = 2.1$ D at 298 K, $x \approx 0.19$ per monomer. Chain segments respond cooperatively over several monomers, so this bias can tip the conformational balance toward β within the bound layer.

4. Dielectric response

The composite permittivity sets both the device capacitance and the field the filler sees during poling. In both roles a large mismatch between ϵ_f and ϵ_m is harmful.

4.1 Effective-medium models

For dilute, isolated spheres the Maxwell–Garnett (MG) expression applies:

$$\epsilon_c = \epsilon_m \frac{\epsilon_f + 2\epsilon_m + 2\phi(\epsilon_f - \epsilon_m)}{\epsilon_f + 2\epsilon_m - \phi(\epsilon_f - \epsilon_m)}$$

We compare it with the symmetric Bruggeman model, the Lichtenecker logarithmic rule and the Yamada model with shape factor n ($n = 3$ recovers MG) [9]. To include the interphase, each particle is replaced by a coated sphere with core fraction $q = [r/(r + t)]^3$:

$$\epsilon_{eq} = \epsilon_i, \frac{(\epsilon_f + 2\epsilon_i) + 2q(\epsilon_f - \epsilon_i)}{(\epsilon_f + 2\epsilon_i) - q(\epsilon_f - \epsilon_i)}$$

The coated spheres are then mixed by MG at volume fraction $\phi(1 + t/r)^3$, capped at 0.74.

4.2 Maxwell–Wagner–Sillars polarisation

Charge accumulates at boundaries between phases with different ϵ/σ ratios. Substituting complex permittivities $\epsilon^* = \epsilon - j\sigma/(\omega\epsilon_0)$ into MG gives a Debye-type relaxation with time constant

$$\tau_{MWS} = \epsilon_0, \frac{(\epsilon_f + 2\epsilon_m - \phi(\epsilon_f - \epsilon_m))}{(\sigma_f + 2\sigma_m - \phi(\sigma_f - \sigma_m))}$$

For a two-layer stack of thicknesses d_1 and d_2 , $\tau = \epsilon_0(\epsilon_1 d_2 + \epsilon_2 d_1)/(\sigma_1 d_2 + \sigma_2 d_1)$.

MWS polarisation has two effects on harvesting. During poling, trapped interfacial charge can stabilise oriented dipoles and add an electret-like polarisation, raising measured d_{33} . In operation, it raises low-frequency ϵ' and loss. That lowers g_{33} and lets generated charge leak away at the 1–10 Hz frequencies typical of body motion.

5. Piezoelectric coupling model

The composite d_{33} is the sum of a matrix term, set by polar crystalline content, and a filler term, attenuated by local field and stress coupling. Signs are kept explicit throughout.

5.1 Constitutive relations and sign

In strain–charge form along the poling axis:

$$D_3 = d_{33}, T_3 + \epsilon_0 \epsilon_{33}^T E_3, \quad S_3 = s_{33}^E, T_3 + d_{33}, E_3$$

Poled PVDF has $d_{33} < 0$. The microscopic origin of this negative sign was analysed by Katsouras et al. [10]. Oxide ceramics and ZnO poled in the same direction have $d_{33} > 0$.

5.2 Local field and stress coefficients

For a dilute sphere in a matrix, Furukawa's local field and stress coefficients are [7]:

$$L_E = \frac{3\varepsilon_c}{2\varepsilon_c + \varepsilon_f}, \quad L_T = \frac{5\varepsilon_f}{3\varepsilon_m + 2\varepsilon_f}$$

Here ε_m and ε_f are the elastic moduli of matrix and filler. L_T tends to 2.5 for rigid spheres in a soft matrix, so stress transfer is favourable. L_E is small when $\varepsilon_f \gg \varepsilon_c$, so field transfer is poor.

5.3 Composite d33

We write the effective coefficient as

$$d_{33}^c = (1 - \phi) d_m(X_c, F_{\text{polar}}) + \phi L_E L_T |d_f|, \\ \quad d_m = d_{\text{ref}} \eta \frac{X_c F_{\text{polar}}}{X_{c,\text{ref}} F_{\text{polar},\text{ref}}}$$

The matrix coefficient scales linearly with polar crystalline content relative to a reference poled film. We take $d_{\text{ref}} = -28$ pC/N, $X_{c,\text{ref}} = 0.50$ and $F_{\text{polar},\text{ref}} = 0.85$; η is the poling efficiency. The sign factor is $s = +1$ for filler poled parallel to the matrix, -1 for antiparallel, and 0 for unpoled filler.

With $s = +1$, the positive filler term opposes the negative matrix term. Ploss et al. showed that poling ceramic and polymer in opposite directions in PZT/P(VDF-TrFE) composites makes their piezoelectric contributions add [8].

5.4 Figures of merit

For off-resonance harvesting, the energy converted per cycle per unit volume scales with the transduction figure of merit:

$$g_{33} = \frac{d_{33}}{\varepsilon_0 \varepsilon_r}, \quad \text{FoM} = \frac{d_{33}^2}{\varepsilon_0 \varepsilon_r}, \quad g_{33} = \frac{d_{33}^2}{\varepsilon_0 \varepsilon_r}$$

Because permittivity appears in the denominator, a filler that raises d_{33} while raising ε_r more can lower the harvested energy.

6. Device model and computational methods

6.1 Electrical model

Below mechanical resonance, a PVDF film acts as a charge source in parallel with its capacitance $C_p = \varepsilon_0 \varepsilon_r A/t$. In compression, $Q = d_{33}F$ and the open-circuit voltage is $V_{oc} = g_{33}\sigma t$. Under a sinusoidal force of charge amplitude Q_0 at angular frequency ω , the average power into a resistive load R is

$$P(R) = \frac{(\omega Q_0)^2 R}{2 \left[1 + (\omega R C_p)^2 \right]}, \quad R_{\text{opt}} = \frac{1}{\omega C_p}, \quad P_{\text{max}} = \frac{\omega Q_0^2}{4 C_p}$$

With an ideal full-wave bridge and diode drop V_d , each half-cycle raises a storage capacitor C_s by $\Delta V = [2Q_0 - 2C_p(V + 2V_d)]/(C_p + C_s)$. The voltage saturates at $V_{oc} - 2V_d$.

6.2 Parameters

Table 3 lists the material parameters. They are representative room-temperature values, not measurements on a specific sample.

Table 3. Material parameters used in the calculations.

Material	ϵ_r	$ d_{33} $ (pC/N)	Young's modulus (GPa)
β -PVDF (poled)	12	28 (sign negative)	2.5
ZnO	8.8	12	140
BaTiO ₃	1700	190	110

The matrix crystallinity is fixed at $X_c = 0.50$ to isolate the effect of polar-phase content. Experiments typically show $F(\text{polar})$ rising with loading and then saturating. We represent this with an illustrative law, $F_{\text{polar}}(\phi) = 0.50 + 0.35[1 - \exp(-\phi/0.03)]$.

The reference device is a 2 cm $\times$ 2 cm, 60 μm film loaded in compression by a 50 N sinusoidal force at 5 Hz.

6.3 Implementation

All models are implemented in the open-source Python package `pvd_f_nano`, with modules for phases, interface, dielectric, piezo and harvester. A `pytest` suite of 32 tests checks limiting cases, Wiener bounds, consistency between the bilayer and sphere MWS expressions, and that numerical maximisation of $P(R)$ recovers R_{opt} and P_{max} . The figures in Section 7 come from the script `examples/design_study.py`.

7. Results

The model predicts that particle size and poling direction matter more to output than the filler's own d_{33} . Figures and tables below use the parameters of Section 6.2.

7.1 Interphase scaling

Interphase share of the composite falls steeply with particle size

Interphase, % of composite volume · 5 vol% filler, 5 nm shell, random-overlap model

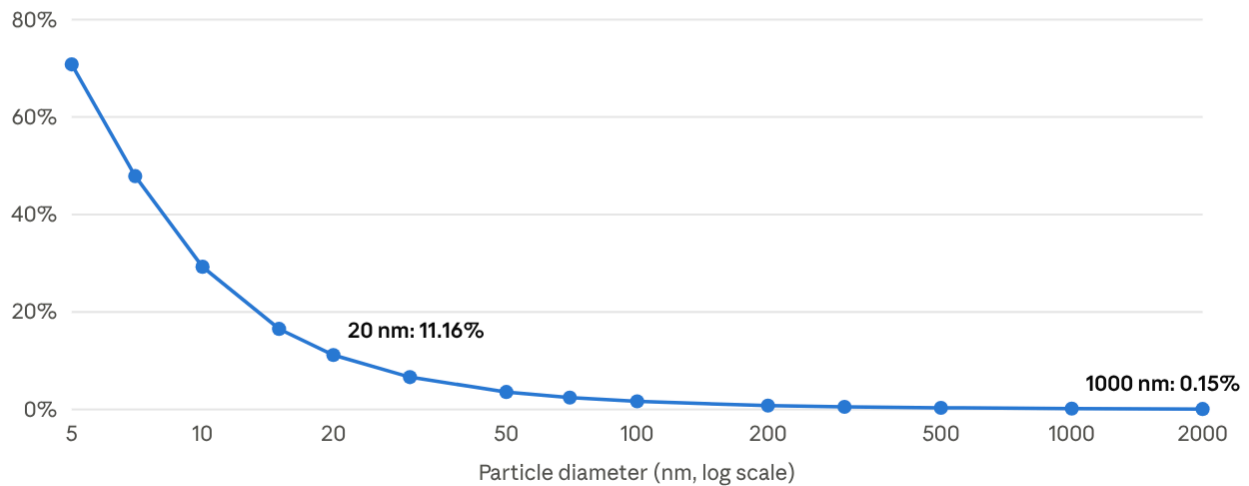


Figure 1 · computed with `pvd_f_nano.interface.interphase_fraction`

The interphase occupies 11.2% of the composite for 20 nm particles, 29.3% for 10 nm particles and 0.15% for 1 μ m particles. At 5 vol%, 20 nm particles sit about 24 nm apart (surface to surface). Their loose layers, tens of nanometres thick, therefore overlap, and most of the matrix behaves as interphase.

7.2 Permittivity

Even at 30 vol% of BaTiO₃, the composite permittivity stays below 70, against about 1700 for the filler (Table 4). The models diverge above about 15 vol%. A 10 nm interphase with $\epsilon_i = 30$ raises the MG value by 11% at 10 vol% and 35% at 30 vol%.

Table 4. Relative permittivity of PVDF/BaTiO₃ (100 nm particles) predicted by four models.

BaTiO ₃ (vol%)	Maxwell–Garnett	Bruggeman	Lichtenecker	MG + 10 nm interphase
10	15.9	17.0	19.7	17.6
20	20.8	28.3	32.3	25.3
30	27.0	69.1	53.0	36.4

The field reaching the filler follows directly. At 10 vol%, $L_E = 0.028$ for BaTiO₃ and 1.09 for ZnO. The stress coefficient L_T is 2.42 for BaTiO₃, so stress transfer is not the bottleneck.

7.3 Piezoelectric coefficient

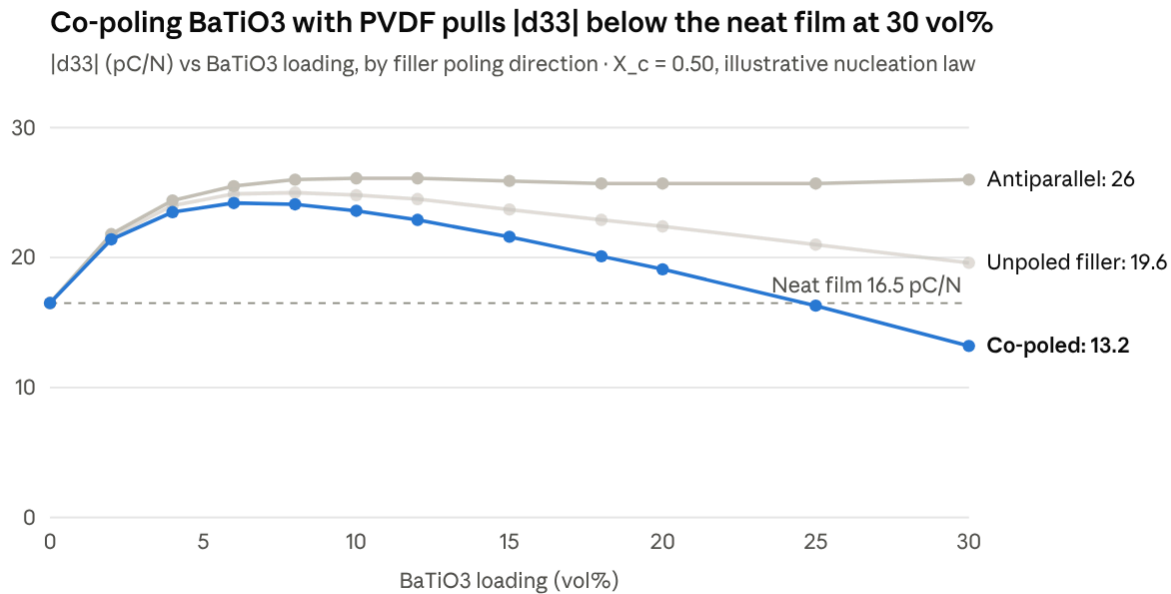


Figure 2 · computed with `pvd_f_nano.piezo.composite_d33`, parameters of Table 3

All three poling cases rise from 16.5 to 24–26 pC/N by 6–8 vol%. This rise comes from polar-phase nucleation, not from the filler. Above about 10 vol% the cases separate. The unpoled case falls to 19.6 pC/N through matrix dilution. Antiparallel poling holds $|d_{33}|$ near 26 pC/N, while co-poling drops it to 13.2 pC/N.

7.4 Device output

At 10 vol% loading and equal polar-phase content, the ZnO film gives the highest open-circuit voltage, figure of merit and power (Table 5). The BaTiO₃ film has the same d_{33} , because its filler term is negligible when unpoled. Its 36% higher permittivity lowers V_{oc} , FoM and power.

Table 5. Predicted output of a 2 cm × 2 cm, 60 μm film under a 50 N, 5 Hz compressive force (filler unpoled).

Film	d_{33} (pC/N)	ϵ_r	$d_{33} \cdot g_{33}$ (10^{-12} m ² /N)	V_{oc} (V)	R_{opt} (MΩ)	P_{max} (nW)
PVDF + 10 vol% ZnO	–24.8	11.7	5.98	1.81	46.3	18
PVDF + 10 vol% BaTiO ₃	–24.8	15.9	4.38	1.32	33.9	13
Neat PVDF ($F_{polar} = 0.50$)	–16.5	12.0	2.55	1.16	44.9	8

Absolute power is tens of nanowatts for this single-layer compression geometry. Bending-mode, multilayer or resonant designs raise it by orders of magnitude, but the ranking between fillers holds.

8. Discussion

8.1 Design rules

The model chain yields five practical rules for PVDF–oxide harvesters.

1. **Use nanoscale, well-dispersed particles.** Interphase volume scales as $1/r$. Agglomerates behave as large particles and add voids that cap the poling field.
2. **Treat surface chemistry as a primary variable.** Hydroxylation, coupling agents and dopants set the sign and density of surface charge, which drives β nucleation.
3. **Prefer low permittivity mismatch.** Fillers with ϵ_f close to ϵ_m (ZnO, KNN) couple to the poling field. BaTiO₃ and PZT need antiparallel or high-field, high-temperature poling to contribute.
4. **Watch the sign.** Co-poling a positive-d₃₃ filler with PVDF reduces the composite response. Report the poling sequence and direction.
5. **Optimise d₃₃·g₃₃ and report test conditions.** State force, area, frequency, thickness and load. Include a polarity-reversal test to exclude triboelectric signals.

8.2 Limitations

The model is first-order. The dilute-sphere coupling coefficients ignore particle interaction, shape anisotropy and percolation. The linear scaling of matrix d₃₃ with $X_c \cdot F(\text{polar})$ neglects crystal orientation and the amorphous contribution to the sign of d₃₃. The nucleation law in Section 6.2 is illustrative and should be replaced by measured FTIR and DSC data. Space-charge electret effects and mechanical resonance are not included.

8.3 Open problems

- Separating nucleation, intrinsic filler response, space-charge effects and triboelectric artefacts in measured d₃₃ and device output.
- Measuring interphase thickness, permittivity and polarisation directly, for example by AFM-IR, piezoresponse force microscopy and Kelvin probe microscopy.
- Accounting for the loss of tetragonality in BaTiO₃ below roughly 50–100 nm.
- Establishing long-term stability of polarisation and space charge under humidity, temperature and 10^6 – 10^8 loading cycles.
- Linking atomistic simulations of surface–chain interactions to crystallisation and continuum electromechanics.

9. Conclusions

In PVDF–metal oxide nanocomposites, the interface controls harvesting performance mainly by setting polar-phase content, not by adding filler piezoelectricity. The interphase occupies about 11% of a composite with 20 nm particles at 5 vol%, against 0.15% for 1 μm particles.

High-permittivity piezoceramics couple poorly. BaTiO₃ sees about 2% of the applied field, and co-poling it with negative-d33 PVDF reduces the composite response. Antiparallel poling reverses this.

At equal polar-phase content and 10 vol% loading, ZnO gives a higher figure of merit than BaTiO₃ (5.98 vs $4.38 \times 10^{-12} \text{ m}^2/\text{N}$) because it raises permittivity less. The released toolkit lets these predictions be tested against measured FTIR, DSC, dielectric and device data.

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